University of Missouri - Columbia

AIAA Design, Build, Fly Proposal

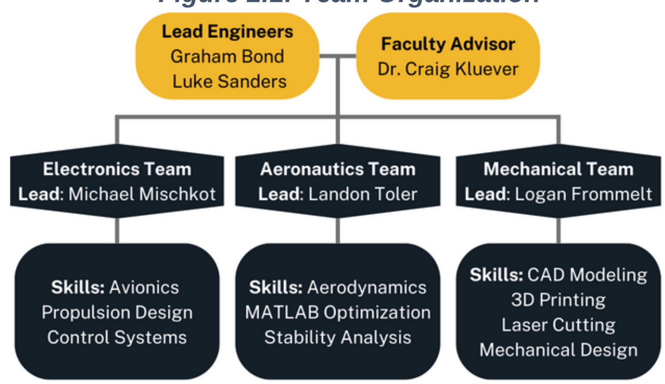
2023 – 2024

1. Executive Summary

This proposal details the approach of the **Mizzou AeroTigers**, representing the University of Missouri – Columbia, to design, analyze, manufacture, test, and fly a scale aircraft at the 2024 AIAA Design, Build, Fly Competition in Wichita, Kansas. This aircraft must be capable of executing missions centered around *Urban Air Mobility*, including a Medical Transport Flight and an Urban Taxi Flight, which require a short takeoff length of 20 ft and the ability to fit in a parking spot with a width of 2.5 ft. After decomposing mission objectives and performing a sensitivity study on design parameters, an optimization program was developed to produce the highest-scoring aircraft. This year's design, **TIGER** (Tomorrow's Infrastructure Gateway and Emergency Rescue), will be a single-motor aircraft with conventional empennage, a high elliptical wing, and the ability to rotate its wing 70° to fit within a parking spot. It will be capable of carrying a Medical Supply Cabinet weighing **6.2 lbs,** completing a lap in **20 seconds** during M2, and will be capable of carrying **60 passengers 9 laps** with a minimum of **20.8 Wh** of stored energy. To execute this design, a project schedule, manufacturing plan, and testing plan have all been developed, based on the key milestones of having a maiden test flight in December, submitting a holistic design report in February, and attending the Fly-Off in April.

2. Management Summary

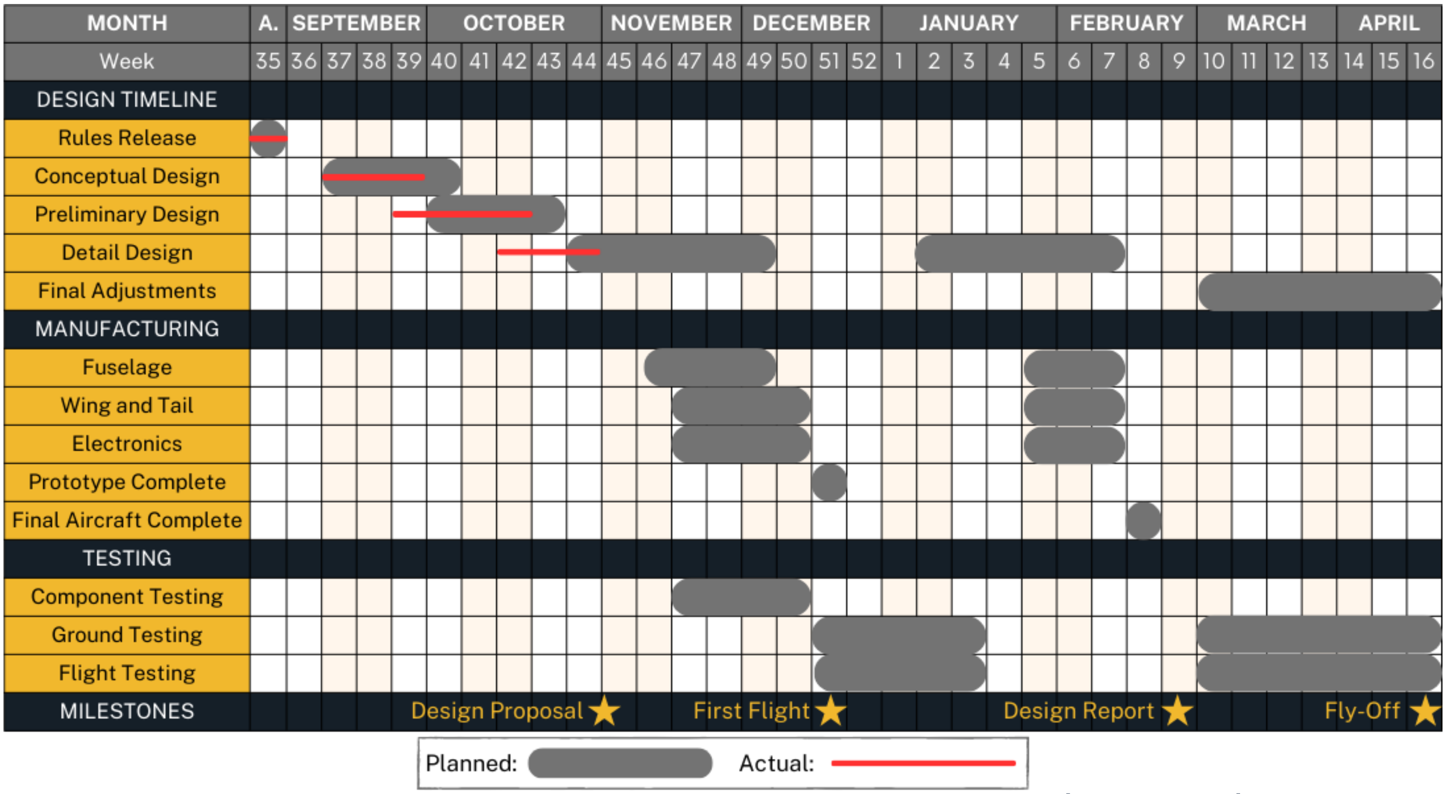
Figure 2.1: Team Organization



2.1 Team Organization: The Mizzou AeroTigers is an entirely student-led organization consisting of three teams: Electronics, Aeronautics, and Mechanical. The teams are organized as shown in **Figure 2.1** and each has a student leader responsible for ensuring the team's timely success. The Electronics team is responsible for the design, testing, and validation of the propulsion, control, and energy storage systems. The Aeronautics team is responsible for the preliminary sizing, planform design, and aerodynamic analysis of the aircraft, which are critical to the optimization of mission deliverables. The Mechanical team is responsible for the design, SolidWorks modeling, and implementation of aircraft structures and mechanisms. All teams are involved in the manufacturing of their respective components and in the writing of this proposal and the design report. Two Lead Engineers supervise and manage the project by creating timetables, integrating the work of the three teams, acquiring funding, recruiting new members, and leading general body meetings.

2.2 Schedule of Project Milestones: Figure 2.2 presents the AeroTigers' timeline to prepare for the Fly-Off. Maintaining scheduled deadlines is key to developing a high-scoring aircraft with ample time for prototyping and testing before the Fly-Off. The AeroTigers' conceptual and preliminary design phases are already complete; section three details these phases and the results. Currently, the detailed design phase is in progress, involving the generation of comprehensive CAD models for manufacturing. Next, prototype manufacturing is scheduled to begin in mid-November, aiming for the first test flight to occur in mid-December. The data and experience gained from each flight will be used to improve the aircraft, with the final design scheduled to be fully manufactured by the end of February. The manufacturing strategy is further detailed in section four, while the testing procedures and planned testing schedule are contained in section five.

Figure 2.2: Competition Gantt Chart



2.3 Budget: The Mizzou AeroTigers' anticipated budget for the 2024 competition is detailed in **Figure 2.3**. The anticipated costs will be financed through a combination of fundraising, corporate outreach, and support from the University. This support will cover manufacturing, testing, and anticipated travel costs. This funding will enable the team to bring 12 students, including leadership and key contributors, to the in-person Fly-Off in Wichita. The AeroTigers have already acquired the necessary manufacturing tools, including 3D printers, soldering equipment, and laser cutters.

Figure 2.3: Budget

CATEGORY	ITEM	ANTICIPATED COST (USD)
Electronics ($1,930)	Motors, Propellers	$450
	Transmitter, ESC	$520
	Batteries	$960
Structures ($1,615)	Carbon Fiber Spars	$1,040
	MonoKote Film	$200
	Landing Gear, Hardware	$100
	Filament/Birch Plywood	$275
Travel ($1,900)	Lodging (4 nights)	$1,400
	Gas (Wichita, KS)	$500
Total Anticipated Cost		**$5,445**

3. Conceptual Design Approach

3.1 Mission Decomposition: This year's challenge is to develop an aircraft specializing in *Urban Air Mobility*. The aircraft must comply with a **5 ft** total wingspan constraint, fit within a **2.5 ft** parking spot, and weigh less than **55 lbs**. Three flight missions will be conducted to gauge the effectiveness of the aircraft design, and one ground mission will evaluate the efficiency of preparing the aircraft for flight from its parking configuration. **Figure 3.1** defines important design parameters and the scoring function for each mission, where $\mathbf{M_n}$ is the individual mission score, **W** is the payload weight, **L** is the number of laps, **P** is the number of passengers, **C** is battery capacity, and **t** is the mission completion time. To compute a total score, the individual mission scores will be summed and multiplied by the design report score, of which 100 points are possible. Flight order is determined by design report score and is paramount to having enough time to complete all missions at the Fly-Off. A small additional score for participation is also included.

Figure 3.1: Mission Decomposition

MISSION	OBJECTIVE	SCORING	DESIGN PARAMETERS
M1: Delivery Flight	• Prepare for flight in under 5 minutes • Complete 3 laps in under 5 minutes	$M_1 = 1.0$	• Ensure takeoff in under 20 ft • Balance CG in unweighted configuration
M2: Medical Transport Flight	• Complete 3 laps in minimum time • Carry heaviest medical supply cabinet • Carry EMTs, Patient on Gurney	$M_2 = 1.0 + \frac{[W/t]_{MU}}{[W/t]_{Max}}$	• Maximize in-flight velocity • Maximize wing strength
M3: Urban Taxi Flight	• Complete maximum laps in 5 minutes • Use minimum battery capacity • Carry maximum number of passengers	$M_3 = 2.0 + \frac{[L^*P/C]_{MU}}{[L^*P/C]_{Max}}$	• Minimize drag for battery efficiency • Balance speed and battery capacity to optimize number of laps • Maximize fuselage storage size
GM: Configuration Demonstration	• Minimize time for inserting and removing each loading configuration	$GM = \frac{t_{min}}{t_{MU}}$	• Design inserts for efficient removal • Design wing for parking configuration

3.2 Sensitivity Study: A sensitivity study was performed in MATLAB to analyze the impact of key scoring and design variables. **Figure 3.2** depicts how each mission objective impacts the scoring function. Lap speed is critical for M2 and M3, and has the largest impact on overall score, while takeoff weight is critical for both M2 payload weight and M3 passenger count. GM time has an inverse, nonlinear relationship with score. M3 Battery capacity has a sawtooth pattern, reflective of the fact that minimum thresholds of battery capacity must be met to complete each lap. Therefore, capacity should be optimized to its minimum necessary value to complete five minutes of laps, as excesses in battery capacity hurt the overall score. Completing less than the maximum number of laps also harms efficiency, as takeoff amperage draw is higher than amperage draw during flight. **Figure 3.3** depicts how wing aspect ratio and takeoff weight impact the overall score at a constant thrust value. Larger aspect ratios decrease the maximum takeoff weight but also decrease drag, allowing higher M2 and M3 flight speeds. The results from the sensitivity study were then used as inputs for generating the optimized preliminary design, with the overall process shown in **Figure 3.4.**

Figure 3.2: Sensitivity of Scoring Variables

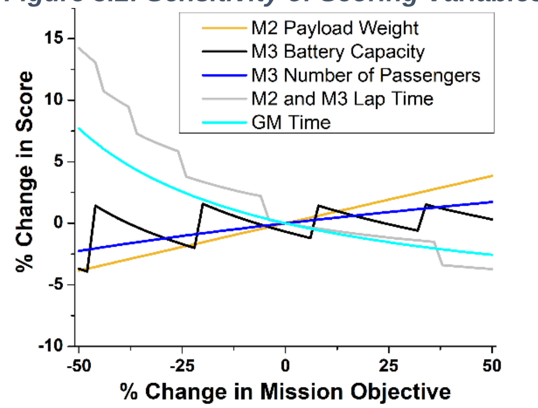
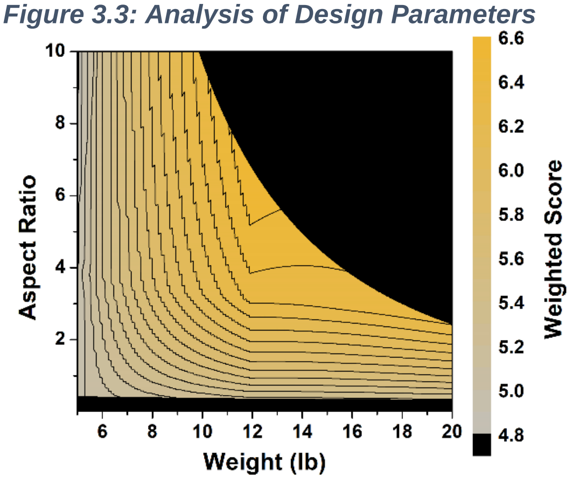
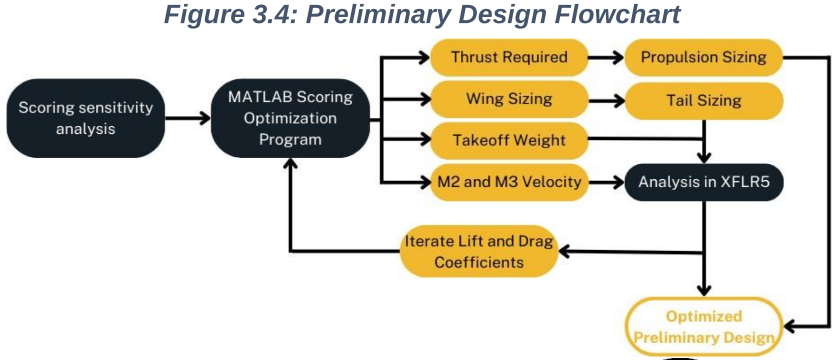


Figure 3.3: Analysis of Design Parameters



3.3 Preliminary Design: Based on the mission decomposition, an effective design must maximize strength, stability, and in-flight velocity while minimizing drag and configuration time. Thus, a monoplane with a high, elliptical wing was selected to optimize the M3 lift-to-drag (L/D) ratio, prevent interference in the parking configuration, and improve stability at a high weight. Tricycle landing gear and conventional empennage were selected

Figure 3.4: Preliminary Design Flowchart



for weight, balance, and ease of manufacturing. The Wortmann FX 60-126 airfoil was selected out of available low-speed airfoils to maximize L/D at the M3 velocity and due to its relatively low thickness.

The preliminary design process was iterative and combined the scoring function generated from the sensitivity study with MATLAB's optimization toolbox to output the highest-scoring aircraft design available. The generated design was then modeled in XFLR5 to output 3D lift and drag coefficients and stability metrics, which were iterated back into the MATLAB program until a converged solution was found. To maximize M3 efficiency, the program included a way to calculate the velocity for maximum L/D and the corresponding thrust required. From this analysis, key results included an aspect ratio of **6.8**, a takeoff weight of **12 lbs**, and a takeoff thrust of **24.4 lbs**. Based on the takeoff weight, a **6.2 lb payload** can be carried in M2 to complete laps at **up to 196 ft/s**, and **60 passengers** can be carried in M3 at an optimized velocity of **80 ft/s**.

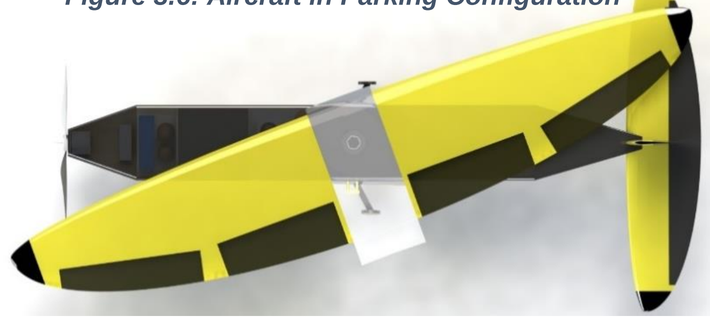
Major restrictions on the propulsion system include the 20 ft takeoff requirement, high takeoff weight, and M3 efficiency. Many motors were compared based on power requirements and amperage draw during takeoff and M3 flight. The **Scorpion HK-4015-1070 kV** was selected, as it can output the required takeoff thrust of 24.4 lb while drawing only **33 A**. In total, an estimated **20.8 Wh** of energy is required to complete all five minutes of M3, equivalent to a 6S 1000 mAh LIPO battery. This motor will be placed in a tractor configuration and will have its propeller size and amperage draw optimized experimentally.

A blended rectangular fuselage was selected to maintain a high fineness ratio while storing the maximum number of passengers. Due to the door width restriction of 6 inches, twelve passenger dolls will be optimally arranged in five individual passenger inserts that are installed through each of the five doors along the side of the aircraft fuselage. To minimize ground mission time, the dolls will be passively held in place by the insert geometry, requiring little assembly effort from the ground crew member. Similar EMT and medical supply inserts will be used for M2. Each insert is shown next to the conceptual model of the aircraft in **Figure 3.5.** The medical supply cabinet will be manufactured from concrete and rebar to achieve the desired weight. Preliminary stability analysis, optimized for M3, has placed an angle of incidence of 1.5° on the wing and -5° on the tail and has located the CG at 35% MAC, creating a static margin of 17% MAC. The M2 and M3 payloads will have their CGs centered in the same place to maintain consistency, while the battery, ESC, and motor will all be placed in the front of the aircraft.

Figure 3.5: Aircraft Conceptual Model

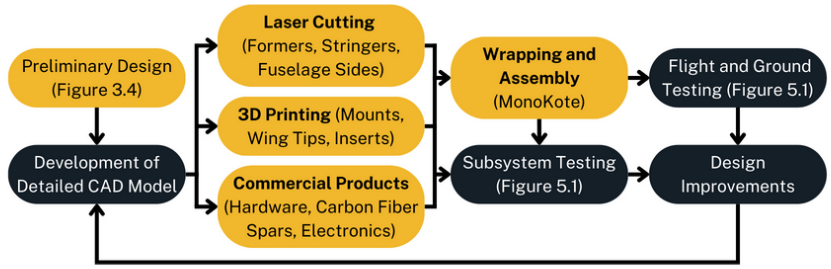
To shift into the parking configuration, shown in **Figure 3.6**, the wing will hinge about a reinforced vertical shaft that connects the fuselage with the primary wing spar. It locks at 0° and 70° and will be held in place with a magnetic key that can be accessed on the fuselage side panel opposite the doors. The carbon fiber wing spars and vertical shaft will be sized based on FEA and experimentation.

Figure 3.6: Aircraft in Parking Configuration



4. Manufacturing Plan

Figure 4.1: Manufacturing Flowchart



The manufacturing flow is outlined in **Figure 4.1**. Weight, strength, and precision all must be considered throughout manufacturing. With a rotating scissor-wing mechanism, the strength of the wing-to-fuselage connection is critical. Each section of the aircraft will be broken down into individual components and modeled in SolidWorks. The primary framework, including ribs, formers, and fuselage sides, will be cut from 1/8-inch plywood using a 60W CO2 laser cutter. More complex components, including the wing rotation mechanism, wing tips, and payload inserts, will be 3D-printed using PLA with the infill optimized for strength and minimal weight. At important locations, including the wing box, wing rotation mechanism, and fuselage, high-strength composite tubes will be used to support the high loads during flight. After each subsystem is manufactured, the aircraft will be assembled with a MonoKote exterior skin. The final competition aircraft will be designed and assembled in the same way with improvements made to the initial design from experience and testing.

5. Testing Plan

Before aircraft assembly, each team will subject their designs to thorough testing and validation. The electronic subassembly will be installed on a test bench to validate the wiring configuration. The propulsion system will be evaluated on a test stand to optimize thrust control and battery discharge rate. Various rotating wing mount prototypes will be loaded to failure to optimize the part's strength and infill volume. Once the first full aircraft prototype assembly has been completed, each structure will be evaluated for strength and performance, with an emphasis on the technical inspection requirements. Ground testing will include a wing tip test, a fail-safe verification, a taxiing assessment, and a measurement of GM time. Flight testing will validate the 20 ft maximum takeoff distance, assess battery endurance, and collect GPS data. M3 velocity will be optimized by comparing throttle input and current draw in flight. The purpose, methodology, and due dates for each test are displayed in **Figure 5.1.** To maintain timely progress, the prototype aircraft will undergo its maiden flight in December and comprehensive testing throughout January. In March, the final aircraft will be subjected to simulated competition runs, requiring the completion of a technical inspection, ground mission, and all three flight missions. Each flight date will allow the pilots to familiarize themselves with the aircraft, fine-tune the control systems, and optimize mission performance.

Figure 5.1: Testing Plan

CATEGORY	TEST	METHOD	GOAL	DUE DATE
Component Testing	Static Thrust Test	Measure thrust, battery usage, and power on a test stand	Select propeller/battery and create thrust model	12/04/23
	Electronics Bench Test	Attach all electronic systems on test bench	Ensure working avionics and propulsion subsystems	12/18/23
	Wing Mount Test	Continuously add weight to rotating wing shaft until failure	Ensure reliability of wing connection during flight	12/18/23
Ground Testing	Landing Gear Test	Drop loaded aircraft onto landing gear	Ensure reliability of landing gear	1/08/24
	Wing Tip Test	Hold each mission-configured aircraft by its wingtips	Ensure structural stability and locate static margins	1/08/24
	Configuration Test	Test GM speed and consistency of changing configurations	Select ground crew and optimize GM score	1/08/24
	Taxi Test	Test movement of aircraft on ground	Ensure landing gear functionality and balance	1/15/24
	Technical Inspection	Perform thorough inspections of the aircraft's construction	Demonstrate ability to pass technical inspection	1/15/24
Flight Testing	Takeoff Distance Test	Measure takeoff distance for different weights and thrusts	Determine throttle required for 20 ft takeoff	1/15/24
	Flight Propulsion Test	Collect data in-flight to correlate thrust with velocity	Generate throttle curve and optimize M3 velocity	1/15/24
	Flight Endurance Test	Measure battery usage in-flight with M3 configuration	Optimize M3 battery capacity and score	1/15/24
	Mock Flight Missions	Practice each flight mission	Ensure stability, optimize M2 velocity and overall score	1/22/24